

Symbolic Category Entropy: A New Entropy Metric for Order-Sensitive Narrative Semantics

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Abstract—Many existing Natural Language Processing (NLP) methods described as *semantic* fail to detect the destruction of meaning when text is shuffled, exposing a discriminant validity gap in computational semantics. We introduce Symbolic Category Entropy (SE or S_Σ), a metric that information-theoretically measures motif concentration. Motifs are defined as semantic categories indexed by certain tokens (words or phrases). Symbolic Category Entropy tracks an important theoretical construct for semantics: the narrative semantic motif structures theorized by Lubomír Doležel. A Shuffle Test is applied to seven widely-used NLP methods and to SE on two datasets, Genesis 1–3 (KJV) and Tolkien’s *The Fellowship of the Ring*. No comparison method achieves Strong detection ($d \geq 1.5$) on sentence-shuffle; SE is the only method to do so ($d = 1.70$, LOTR), and the only method achieving Strong on both word-shuffle and sentence-shuffle simultaneously. Symbolic Category Entropy is quantified via Cohen’s d on the per-window ℓ_2 -norm of the S_Σ KL-contribution vector and visualized using heatmaps. SE consistently shows distinguishable outputs that contrast original, word-shuffled, and sentence-shuffled versions of corpora. This demonstrates order-sensitive semantic measurement where all seven comparison methods fall short.

Index Terms—symbolic entropy, narrative semantics, shuffle test, NLP evaluation, order sensitivity, computational semantics, information theory, motif analysis, text indexing and retrieval

I. INTRODUCTION

Although methods called *semantic similarity* and *semantic analysis* are widely used in NLP, they primarily capture statistical co-occurrence rather than structural meaning. When we submitted a range of popular methods to a falsification test designed to assess order-sensitivity, the results were stark. Collectively, these methods capture either local syntax (perplexity), statistical co-occurrence (topic models), or order-independent features (sentiment, NER, BERTScore), revealing a gap in sensitivity to higher-order semantics among many major methodologies. Of the tested methods, only SE reliably detects both word-order and sentence-order disruption strongly and simultaneously. If some methods reflect sensitivity to features of a corpus rather than the shuffle, a lack of methods sensitive to higher-order narrative structure is strongly indicated.

We argue that a true semantic analysis method must capture the order-sensitive meaningful phenomena occurring in a text, such as the order in which key thematic structures occur. When submitted to a Shuffle Test, the analysis should differ sharply. In other words, when applied on an order-randomized sample,

the output should be significantly different from when applied to the original, thus indicating a detection of the shuffle’s destruction of meaningful patterns. For example, singing a song in order does not have the same meaning as singing the lyrics in a random configuration. However, numerous methods currently labeled *semantic* cannot clearly tell the difference.

This paper introduces Symbolic Category Entropy (SE), a KL divergence measure that quantifies the concentration of motif category surprise. Claude Shannon explicitly left open the question of measuring semantic qualities in information, writing that the “semantic aspects of communication are irrelevant to the engineering problem” [1]. SE analysis can plot Shannon entropy, which measures lexical complexity, over SE, which measures how far the concentration of motifs deviates in a section from their average distribution across the full text. When an ordered text is shuffled, its intended motif structure is destroyed; SE detects this destruction while many existing methods cannot.

The contributions of this paper are as follows: (1) We repurpose the Shuffle Test as a criterion for evaluating the order-sensitivity of NLP methods, (2) the Shuffle Test is applied to seven widely-used methods and results demonstrate how they respond to the destruction of order-sensitive meaning, (3) we apply Symbolic Category Entropy (SE or S_Σ) as an information-theoretic methodology for tracking narrative semantic structure, and (4) we demonstrate that SE strongly distinguishes original from shuffled text across two corpora.

II. RELATED WORK

A. Shuffle Testing in NLP

The Shuffle Test rests on a simple intuition: if a method measures semantic structure, it should have some means of reliably distinguishing a properly structured text from randomized versions of it. Compositional semantics as well as basic common sense tells us that a text’s meaning depends on the arrangement of its parts. If a page’s sentences and words are scrambled, meaning is destroyed. A metric that fails to detect this destruction cannot be measuring meaningful structure; it can only be said to measure some kind of meaningful content [2].

The Shuffle Test’s intellectual ancestry begins in cognitive psychology. Kintsch and van Dijk’s “macrostructure theory” showed that readers organize texts into hierarchical levels

using internalized genre schemas [3]. Kintsch, Mandel, and Kozminsky demonstrated that readers could mentally reorganize scrambled narratives with strong causal structure [4]. Grosz and Sidner extended this computationally, proposing that discourse coherence emerges from interrelated linguistic, intentional, and attentional structures [5]. Barzilay and Lapata’s pivotal entity grid model converted texts into a matrix (sentences $\times$ entities) which tracked the presence of entities in sentences. They used a Shuffle Test to validate that coherent texts exhibit systematic regularities in entity recurrence [6]. In 2011, Micha Elsner and Eugene Charniak extended the entity-grid framework with linguistic and entity-specific features [7]. Their work represented Shuffle Testing’s most refined use before neural approaches came to dominate the field.

B. The Benchmark Gap

The entity grid’s decline coincided with the rise of word embeddings. Word2Vec demonstrated that high-quality vector representations could be trained on billions of tokens in hours, but it was built on an order-insensitive architecture. Mikolov and colleagues acknowledged that “an inherent limitation of word representations is their indifference to word order,” [8]. Shuffle Testing did not fail; it became structurally incompatible with how success was defined [9]. Output-based standards created silent discriminant validity problems as methods were optimized for accuracy rather than semantic sensitivity. However, a recent revival has begun. Abdou et al. [10] demonstrated how Sinha et al.’s claim that shuffled-text training barely hurt performance rested on allowing order information, via positional encodings and statistical regularities, to leak into supposedly orderless data [11]. When corrected, shuffled models showed severe degradation on tasks requiring structural understanding. Chen et al. [12] offered an information-theoretic explanation: models tend to behave consistently only when scrambled sentences still preserve enough mutual information to recover the original order. The benchmark era had not proven that word order was dispensable but had optimized for tasks where it happened to be redundant. Similarly, Laban et al. [13] showed that a fine-tuned model achieved 97.8% accuracy on the Shuffle Test not by learning coherence, but by exploiting surface features specific to shuffled text.

C. Narrative Semantics and Motif Theory

Contemporary semantic theory operates through distinct but overlapping domains: lexical semantics analyzes word-level meanings, compositional semantics derives sentential meanings through truth-conditional operations, pragmatics examines context-dependent implicatures, discourse semantics investigates coherence relations, and cognitive semantics explores conceptual structures (e.g. frames and metaphors) [14]. These frameworks address how meanings combine and how context enriches content but they share a common limitation: lacking a notion for understanding semantic patterns that extend across entire texts and recur across multiple texts.

Consider the various global flood myths. The Epic of Gilgamesh, Genesis, Indian and Chinese flood narratives, and

numerous Native American stories all preserve structurally similar motifs of catastrophe, survival, preservation, and renewal. These parallels do not depend on shared vocabulary, identical syntax, or the same pragmatic context. Rather, their commonality appears at the level of narrative event-structure: a patterned sequence in which destruction, selection, preservation, and re-creation are temporally organized. Semantic categories may describe aspects of these stories, but do not sufficiently explain why the same symbolic structure remains recognizable across languages, discourses, and contexts. This suggests the need for a higher-order semantic account of motif persistence across narrative instances.

Doležel formalized this dimension in 1978 as narrative semantics, establishing that texts operate through underlying motif structures indexed by surface words. His framework identifies motifs as “the basic classes of elementary semantic units of narrative” [15]. His analysis of the “Natural Force” motif demonstrates the core principle: rarely does an expression corresponding to the motif appear directly in a text’s surface structure. Instead, specific tokens (“lightning,” “storm,” “wind,” “sea”) function as what Doležel terms “semantic instruments” for the underlying motif (Natural Force). This defines the motif category for SE as well: a motif is an overarching concept indexed by a set of words or phrases. Multiple surface tokens can index the same motif because they share semantic properties. The question that defines narrative semantics is: “What broad trans-textual meaning patterns have been made present here?” This question is not adequately addressed within existing computational frameworks.

The idea that meaning exists through a dimension of universal or widely shared patterns has a long history in philosophy and the study of religion, from Platonic forms to cross-cultural patterns as documented by scholars like Marie-Louise von Franz [16] and Mircea Eliade [17]. The Swiss psychiatrist Carl Jung even postulated the existence of a repository of narrative archetypes shared by humanity in a “collective unconscious,” [18]. SE does not depend on any particular metaphysical commitment regarding the ultimate nature of archetypes or motifs; rather, it operationalizes their measurement while leaving ontological questions open.

Cyber Humanities is defined by Bellini and Adorni as a “foundational paradigm for humanistic inquiry in a computationally mediated world” that integrates algorithmic methods with humanistic questions about meaning, culture, and memory [31]. This provides the disciplinary context for SE. Within this paradigm, computational approaches to tracking thematic patterns in literary texts have a substantial history. Moretti’s “distant reading” paradigm proposed analyzing literature through aggregate quantitative features [19]. Jockers extended this with macroanalysis, using word-frequency signals to trace thematic trends across thousands of novels [20]. These approaches demonstrate the value of computational pattern-tracking but rely on raw word frequencies or topic model outputs without detecting the patterns of ordered semantic structure. SE differs in three respects: (a) KL divergence normalizes local motif counts against a global baseline, con-

verting raw frequencies into information-theoretic surprise values measured in bits per token; (b) a two-component design couples lexical surprise (H) with motif concentration surprise (S_Σ), capturing dimensions of meaning that frequency counts alone miss; and (c) the Shuffle Test demonstrates SE’s order-sensitivity.

III. METHODOLOGY

A. The Shuffle Test as a Falsification Criterion

The Shuffle Test can be used to assess the effectiveness and output type of a semantic analysis method. Our hypothesis is that if a metric can measure some order-sensitive aspect of meaning, its value should change significantly when comparing unaltered text with a different version where word or sentence order is randomized. The following Algorithm 1 formalizes this test [2].

Algorithm 1 Shuffle Test for Order-Sensitivity

Require: Text T , Metric M , Number of iterations N

- 1: Compute $M(T_{\text{original}})$
- 2: **for** $i = 1$ to N **do**
- 3: Generate new random seeds
- 4: Generate $T_{\text{word-shuffled}}$ (randomize word order)
- 5: Generate $T_{\text{sent-shuffled}}$ (randomize sentence order)
- 6: Compute $M(T_{\text{word-shuffled}})$ and $M(T_{\text{sent-shuffled}})$
- 7: **end for**
- 8: Compute Cohen’s d between $M(T_{\text{original}})$ and a shuffled distribution

Ensure: Strong: $d \geq 1.5$; Moderate: $0.8 \leq d < 1.5$; Weak: $0.5 \leq d < 0.8$; Fail: $d < 0.5$

Passing the Shuffle Test means that a method produces a significantly different score on a shuffled text compared to its original, signaling that it is sensitive to narrative order. Failing the test means registering comparable semantic content in the original and the shuffle, for example, a topic model finding topics in shuffled text despite order disruption causing no meaningful topics to be present.

B. Symbolic Category Entropy: Definition

The novel contribution is visualized with Shannon Entropy (H). This H-Overlaid SE model (HOSE) is a two-component demonstration designed to capture narrative semantic data. HOSE shows how both Shannon Entropy (H) and SE/Symbolic Category Entropy (S_Σ) perform in a given text window. While the standard methodology involves only the heatmap, overlaying H gives a more complete picture of how a text performs in both lexical diversity and thematic concentration. This is computed as follows:

$$\text{HOSE} = (H, S_\Sigma) \quad (1)$$

where H is calculated per window using the standard Shannon entropy formula:

$$H = - \sum_i p(x_i) \cdot \log_2 p(x_i) \quad (2)$$

Here $p(x_i)$ is the probability of word x_i in a given window. Higher H indicates more diverse, unpredictable vocabulary; lower H indicates repetitive, predictable language. S_Σ is calculated per window and then compared with the global baseline via KL divergence:

$$S_\Sigma = \sum_{m=1}^k p(m) \cdot \log_2 \frac{p(m)}{q(m)} \quad (3)$$

where the summation is over all k user-defined motif categories, $p(m)$ is the local motif distribution within the window, and $q(m)$ is the global baseline distribution across the entire text. If “power” appears in 1% of the full text but 8% of a single window, that window produces a high S_Σ value, indicating that the section contains a high motif surprise concentration. The result is measured in bits per token and indicates, via KL divergence, how much the window concentrates motif patterns beyond the baseline global average. H is calculated in each window and globally, without comparison to a baseline.

C. Procedure

The H-Overlaid SE methodology proceeds in six steps and some example visualization results are depicted on page 5.

Step 1: Tokenization. Take the text and merge any multi-word phrases defined in the motif set into single tokens. For example, if “Ring of Power” indexes a motif, all occurrences are merged into “ring_of_power.” Then split the rest of the text into individual tokens: “The Fellowship of the Ring” becomes “the,” “fellowship,” “of,” “the,” “ring.” Both single words and merged phrases are treated as tokens in all subsequent steps, and the total token count per window reflects this combined token stream.

Step 2: Compute Shannon Entropy (H). For each window, calculate the probability of each unique word by dividing its count by the total number of words. For each unique word, multiply its probability by the binary logarithm of that probability. If “ring” appears 3 times in a 500-word window, $p(\text{ring}) = 0.006$, yielding $-0.006 \times \log_2(0.006) = 0.044$ bits. Sum all bits/token results for each window and for the whole text to get H per window and global H .

Step 3: Build a Motif Dictionary. Identify around 100–500 frequent and meaningful tokens in the text, including both individual words and multi-word phrases, then group them into 10–30 categories that capture the text’s thematic structure. This is a rough estimate that should be liberally adjusted based on scope and corpus size. For a fantasy epic, categories might include FELLOWSHIP (companion, friend, ally, together), SHADOW (darkness, evil, enemy, Mordor), JOURNEY (path, road, quest, wander), POWER (ring, master, rule, dominion). The categories should feel natural to the text’s meaning. This is where human interpretation enters the process.

Step 4: Create Sliding Windows. Split the text into overlapping chunks. Window size can be chosen based on what is best for the volume of the text; for the texts in this study,

an adaptive window size is computed targeting approximately 80–120 windows across the text with a 50% fixed overlap.

Step 5: Compute Symbolic Category Entropy (S_Σ). For each window, count how many tokens belong to each motif category and convert to probabilities by dividing by the window size, yielding $p(m)$, the local motif distribution. Then calculate $q(m)$, the global baseline, from the full text. Apply Equation 3 to compute S_Σ for each window by summing over all motif categories.

Step 6: Plot HOSE = (H, S_Σ) . Heatmaps best visualize the results. High positive values are indicated by brighter coloration and point to motif presence in bits/token beyond global average expectation; negative values (per motif) indicate below-expectation presence; zero indicates zero symbolic category entropy/surprise (the motif is absent or present at baseline). In the case of H-Overlaid SE, Shannon entropy is overlaid as a line plot to show how lexical complexity changes in relation to motif presences.

Table I shows a 19-category motif dictionary built for the Genesis 1–3 corpus used in our experiments. Each category groups semantically related words and multi-word phrases that index the same underlying narrative motif. The dictionary contains 253 words and 100 phrases across 19 categories, identifying the most frequent and thematically significant tokens, grouping them according to the narrative structures present in the passage.

TABLE I
MOTIF DICTIONARY FOR GENESIS 1–3 (KJV): 19 CATEGORIES

Category	Words	Phrases
Tree / Axis Mundi	17	4
Waters / Sea	20	4
Light	25	4
Sharpness / Sword / Thorn	4	2
Death / Mortality	3	6
Eating / Forbidden Fruit	6	6
Curse / Toil	4	4
Transgression / Deception	2	5
Desire / Wisdom	4	6
Fear / Hiding	2	6
Exile / Expulsion	3	4
Serpent / Beast	24	6
Bridegroom–Bride	29	7
Garden / Paradise	25	4
Clothing / Nakedness	15	4
Sacred Name / Word	31	11
Good–Evil / Knowledge	15	2
Command / Obedience	8	6
Time / Order	16	9
Total	253	100

D. Interpreting the Two Components

The H-overlaid method finds places where meaning concentrates through the interaction of surprise and pattern. Together, the two components (H and SE) capture how a text fares in both lexical diversity and thematic concentration. SE deliberately produces a two-dimensional output because narrative

meaning is temporally distributed across a text; collapsing motif concentration patterns into a single scalar would discard the structural information that distinguishes SE from existing approaches.

One might object that applying scalar metrics to competing methods while evaluating SE through distributions constitutes an asymmetric standard. But the asymmetry is in the methods, not the evaluation. The comparison methods were designed to produce scalars and are evaluated on their own terms. Using scalars for SE removes its key design feature of capturing distributions. A 2D visualization enables us to determine if a text’s structure is distinguishable between shuffle conditions, which Figures 1 and 2 clearly depict on page 5. The ℓ_2 -norm scalar captures only total motif-concentration magnitude per window; it cannot encode which categories co-activate, nor whether clustering is spatially coherent across the narrative, but these are key dimensions, visible in the S_Σ heatmap, that are destroyed by shuffling. SE passes the Shuffle Test simultaneously along multiple orthogonal axes of measurement: per-category activation, cross-category co-occurrence, and temporal coherence of clustering. The ℓ_2 -norm scalar is a single-axis projection; SE’s signal is multi-dimensional, and the heatmap is an appropriately comprehensive representation.

IV. SHUFFLE TEST EXPERIMENTAL RESULTS

A. Experimental Setup

We compared SE with seven widely-used NLP methods (each representing a different approach) on Tolkien’s *The Fellowship of the Ring* and Genesis 1–3 in the King James Bible (181,000 and 2,000 words respectively). The texts differ in length, motif structures, and style. Each method was applied to: (1) the original text, (2) a version with the word order shuffled, and (3) a variant with shuffled sentence order. The effect sizes were computed using Cohen’s d . The open-source Shuffle Testing framework and SE calculator are available on GitHub [2], [21].

B. Shuffle Test Results

Table II presents the results of the Shuffle Test by method.

TABLE II
SHUFFLE TEST RESULTS: COHEN’S d BY METHOD

Method	Genesis Word		Genesis Sent		LOTR Word		LOTR Sent	
	d	Rating	d	Rating	d	Rating	d	Rating
Perplexity	5.75	Strong	1.46	Mod	7.56	Strong	0.88	Mod
LDA	3.46	Strong	1.32	Mod	1.36	Mod	0.73	Weak
BERTopic	1.23	Mod	1.30	Mod	0.37	Fail	0.35	Fail
TF-IDF	0.91	Mod	1.04	Mod	0.79	Weak	0.73	Weak
BERTScore	0.28	Fail	1.17	Mod	3.59	Strong	1.03	Mod
Sentiment	0.43	Fail	0.23	Fail	0.97	Mod	0.16	Fail
NER	1.02	Mod	0.88	Mod	0.06	Fail	0.01	Fail
SE	1.93	Strong	1.07	Mod	1.89	Strong	1.70	Strong

Three-Way Comparison: Hierarchical Structure Destruction

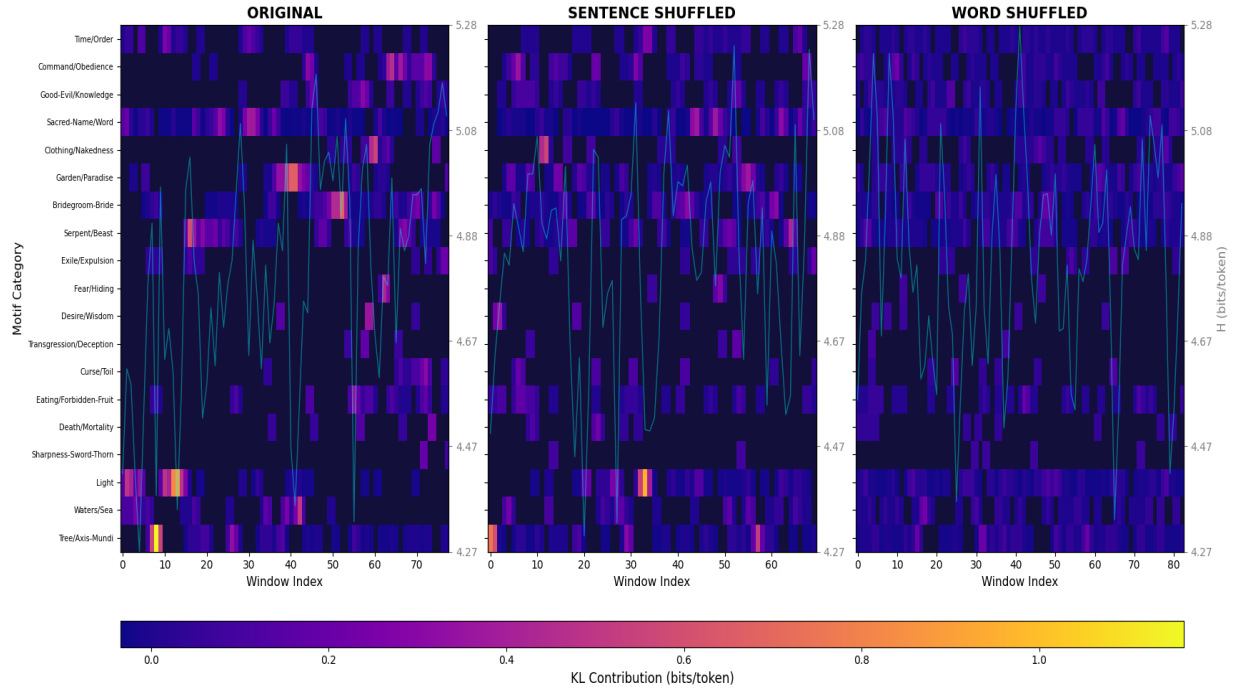


Fig. 1. SE heatmap comparison of Genesis 1–3 (KJV). Left: original text shows structured motif clusters corresponding to narrative segments. Center: word-shuffled text collapses to near-baseline values with no detectable structure. Right: sentence-shuffled text shows randomly distributed spikes with no coherent clustering. 19 motif categories (see Table I) were used.

Three-Way Comparison: Hierarchical Structure Destruction

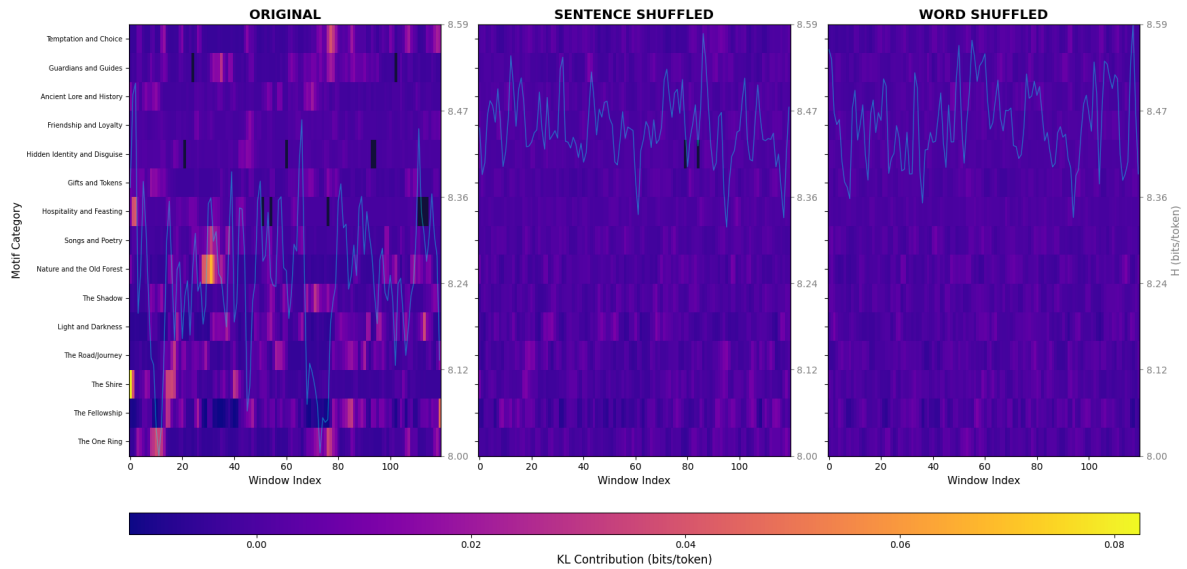


Fig. 2. SE trifold heatmap for *The Fellowship of the Ring* (Tolkien). Left: original text shows structured motif clustering across narrative events. Center: word-shuffled text shows collapse of motif concentration toward baseline. Right: sentence-shuffled text redistributes spikes without coherent clustering. 15 motif categories tracked across 100-150 windows. The Overlaid H (cyan) line plot shows how lexical complexity interacts with motif surprise across the duration of the narrative.

The results reveal dimensions of semantic sensitivity.¹ Perplexity achieves strong word-shuffle detection in both texts (LOTR $d = 7.56$, Genesis $d = 5.75$) because destroying token order spikes GPT-2 perplexity 30–50 $\times$, while sentence-shuffling reaches only moderate detection since within-sentence grammar remains intact. LDA achieves strong word-shuffle detection on Genesis ($d = 3.46$) and moderate on LOTR ($d = 1.36$) likely through topic-concentration collapse despite being order-invariant. Sentiment achieves Moderate detection on LOTR word-shuffle ($d = 0.97$) but Fails on three of four conditions. BERTopic, despite relying on order-sensitive positional encodings, fails on LOTR ($d = 0.37, 0.35$) but reaches moderate on Genesis ($d = 1.23, 1.30$). This is for the same reason as LDA: Genesis’s day segments produce concentrated topic assignments that collapse under shuffling, a handle absent in LOTR’s uniform narrative. NER detects word-shuffling only on Genesis ($d = 1.02$) and fails on LOTR ($d = 0.06$), possibly due to how named entities are defined in training.

In Genesis, TF-IDF and BERTScore perform better under sentence d because word-shuffling homogenizes consecutive windows into similar vocabulary distributions while sentence-shuffling creates genuine lexical discontinuities. Among all seven comparison methods, none achieves Strong detection ($d \geq 1.5$) on sentence-shuffle; SE is the only method to do so with the LOTR corpus ($d = 1.70$, LOTR). It is also the only method achieving Strong on both word-shuffles and LOTR’s sentence-shuffle. Thresholds for strength follow Cohen (1988) and Sawilowsky (2009) [29], [30]. Cohen’s levels (small $d = 0.2$, medium $d = 0.5$, large $d = 0.8$) were calibrated for behavioral-science experiments; complete structural interventions operate at a different scale. Sawilowsky’s extended taxonomy adds “very large” ($d = 1.2$) and “huge” ($d = 2.0$). We define Strong at $d \geq 1.5$, between these two upper levels, confirmed by a natural gap in the data whereby most results cluster between 0 and 2. Results that greatly exceed this range appear to be highly sensitive to order disruption.

C. SE Performance

SE distinguishes original from shuffled text across both shuffle types and corpora. Figure 1 presents three SE heatmaps of Genesis 1–3: the original text, a word-shuffled variant, and a sentence-shuffled variant. The original text displays structured

motif concentrations corresponding to recognizable narrative segments. “Sacred Name/Word” and “Light” concentrate in the earliest windows where God names and divides the elements of creation. “Transgression/Deception,” “Fear/Hiding,” and “Exile/Expulsion” spike together at the Fall and its aftermath. Word-shuffled heatmaps collapse to a flat field of near-zero values. Sentence-shuffled heatmaps distribute peaks randomly without coherent clustering. The difference is obvious to any viewer, regardless of their familiarity with the source text. SE passes the Shuffle Test when the original heatmap displays localized motif clusters corresponding to known narrative segments that are absent in shuffled variants. The structured-versus-flat distinction visible in Figure 1 leaves no ambiguity about whether narrative order is present.

A positive KL divergence means that a motif is thematically prominent and occurs with positive surprise in excess of its average expectation. A negative KL for an individual motif means the motif is operating as an undertone, present but below its usual intensity. A value of zero means the motif is genuinely absent or present at exactly its global rate. SE captures these three distinct modes of semantic operation in a single information theoretic measurement. It also maps onto how literature actually works. Tolkien doesn’t stop mentioning darkness when writing about other themes — he keeps it murmuring underneath so the reader feels unease without necessarily knowing why. The LOTR results confirm SE’s applicability to a diverse, longer corpus. NER, TF-IDF, and BERTScore each perform differently across the two texts, indicating their sensitivity depends on textual properties rather than methodology form alone.

Figure 2 presents the SE Shuffle Test on *The Fellowship of the Ring* across 15 motif categories and over 100 sliding windows. In the first heatmap focusing on the original, a narrative structure is clearly visible: “The Shire” concentrates sharply in the opening windows and vanishes once the hobbits depart, appearing only when reflected upon nostalgically; “Nature/Forest” clusters at the Old Forest and Tom Bombadil chapters, and “The One Ring” spikes at Bilbo’s party, the Council of Elrond, and around the breaking of The Fellowship. The word-shuffled heatmap collapses this structure, with S_Σ values flattening toward baseline. The sentence-shuffled heatmap also collapses and large, ordered motif spikes vanish. More so than with Genesis 1–3, in part due to the much higher token count, the crash in S_Σ upon both shuffles is drastic and unambiguous.

To provide a scalar summary comparable to Table II, we report Cohen’s d on the per-window ℓ_2 -norm of the S_Σ KL-contribution vector. Three findings emerge. First, SE and Perplexity are the only methods achieving Moderate or above across all four conditions; every other method has at least one Weak or Fail. Second, SE is the *only* method achieving Strong on any sentence-shuffle condition ($d = 1.70$, LOTR), exceeding Perplexity’s LOTR sentence-shuffle ($d = 0.88$) by 0.82 Cohen’s d points. Third, SE is the only method achieving Strong on *both* word-shuffle and sentence-shuffle simultaneously. These results hold when the 2D heatmap is

¹Perplexity was computed using GPT-2 [22]; results may vary slightly across hardware and PyTorch versions due to floating-point non-determinism. Sentiment analysis used DistilBERT [23] on 80-word chunks. TF-IDF coherence [24] measured cosine similarity between consecutive 50-word units. NER density was measured with spaCy [25] as entity count divided by 50-word unit. LDA topic modeling used Gensim [26], retaining maximum topic probability per 50-word unit. BERTScore [27] computed consecutive similarity on 50-token windows. BERTopic [28] retained maximum topic probability per 50-word unit. All Cohen’s d values used pooled standard deviations. LDA’s variational inference is seeded at 42 but remains sensitive to floating-point execution order across hardware and Gensim versions, meaning results may differ slightly across environments despite the fixed seed. BERTopic’s HDBSCAN clustering step uses no fixed seed, producing slightly different cluster shapes, topic assignments, and keyword lists across runs; topic assignments and resulting Cohen’s d values will thus vary. Results reported here represent a single sample and are indicative rather than fixed.

collapsed to the per-window ℓ_2 -norm scalar; the full heatmap measurement is all the more discriminating.

V. DISCUSSION AND FUTURE WORK

A. Limitations

Some limitations of the SE methodology include: (1) fully automated motif discovery remains an open problem. Motifs are currently manually defined and fine-tuned by a human or an LLM; (2) accuracy of the analysis needs to be validated to ensure that any automated process aligns with human understanding of how meaning is moving through a text; (3) experiments relied on two texts of different genres and sizes. As a result, further validation on diverse text types, languages, and domains is necessary to generalize this method. Since the concept of *meaning* inevitably includes subjective judgment, we allow for different modes of categorization based on user-defined interests. Therefore, SE analysis can be applied on the same text in different ways. For example, one analysis can focus on emotional motifs, another on specific character dynamics, or on any combination of meaningful token categories.

B. Motif Validity

The claim that motif curation is purely arbitrary is untrue. Less arbitrary motif selections should collapse harder under shuffling. This is because motif categories that best track what is actually happening in the discourse structure will show clusters and S_Σ spikes significantly more than an arbitrary set of motifs. Crash magnitude could serve as a potential metric for testing motif-set accuracy.

C. Implications

Methods that cannot distinguish ordered from disordered text are not measuring meaning in a sense that satisfies philosophical, literary-theoretical, or commonsense criteria for what “semantic” means. This has implications for NLP pipelines in meaning-sensitive domains: legal document analysis requires sensitivity to how case law develops; clinical narrative processing must track how symptoms emerge over time; AI-assisted literary scholarship depends on detecting narrative structure. In each domain, methods that pass the Shuffle Test are a minimum requirement. The Shuffle Test itself should serve as a validity check for any method claiming narrative semantic sensitivity. If motif concentration is measurable in bits per token, this opens avenues in the digital humanities, and more generally in any domain in which *ordered informational structures* matter, such as genetics, music, and AI attention mechanisms.

D. Future Work

The methodology is extensible. Sentiment analysis can supply emotional motifs, topic modeling can reveal hotspots of motif activity, and Symbolic Category Entropy analysis can be informed by other methods and technology.

Additionally, automating SE becomes easier in domain-specific applications. For example, a therapy session centers

on specific patterns, words, and phrases so motif-sets can be trained to categorize token data accordingly. Different data types eliminate the arbitrariness problem altogether. Music already occurs in sequences of motifs such as note names, ranges, and dynamics. Bioinformatic data organizes content according to objective motifs like genetic sequences. Thus, SE could be used to track motif patterns in different kinds of data, not just texts.

VI. CONCLUSION

What SE accomplishes is not the discovery of narrative semantics but its first information-theoretic operationalization. For millennia, thinkers kept encountering the same phenomenon: meaning that emerges from pattern-structure, not just word-combination. They described it philosophically and analyzed it in literary theory. But they could not measure it because the mathematics did not exist yet. Shannon gave us H and deliberately left the semantic problem to future engineers. Kullback and Leibler gave us distributional comparison. The Shuffle Test gives us falsification. We introduced Symbolic Category Entropy (SE) and demonstrated that it consistently distinguishes original from shuffled text across two corpora of different genres and sizes. SE achieves Cohen’s $d \geq 1.89$ (on the per-window ℓ_2 -norm of the S_Σ KL-contribution) (Strong) on word-shuffle for both corpora and $d = 1.70$ (Strong) on LOTR sentence-shuffle. SE is thus the only method in our comparison to achieve Strong on any sentence-shuffle condition and the only method achieving Strong on both LOTR word-shuffle and sentence-shuffle simultaneously. H-Overlaid SE’s two-component design captures both lexical diversity and thematic concentration in a single framework measured in the common unit of bits per token, and its heatmap outputs make the presence or absence of narrative structure legible at a glance. In taking up the project Shannon deferred, SE operationalizes Doležel’s narrative semantics through information-theoretic concepts and opens new avenues for computationally analyzing meaning in the digital humanities, data science, and beyond. The strength of an S_Σ measurement depends on the strength of its motif dictionary as a theory of meaningful patterning. The code for the SE calculator, heatmap visualizer, and Shuffle Testing framework is open-sourced and available for replication.

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